

When the comparator population changes

Does a pathway priority survive a change in which perturbations are analyzed?

Public A549 WT / PRDX1-KO × vehicle / AZD7648 screen

Fixed mitochondrial and translation-elongation sets

Finding: the comparator changes; the mitochondrial component is nearly stable.

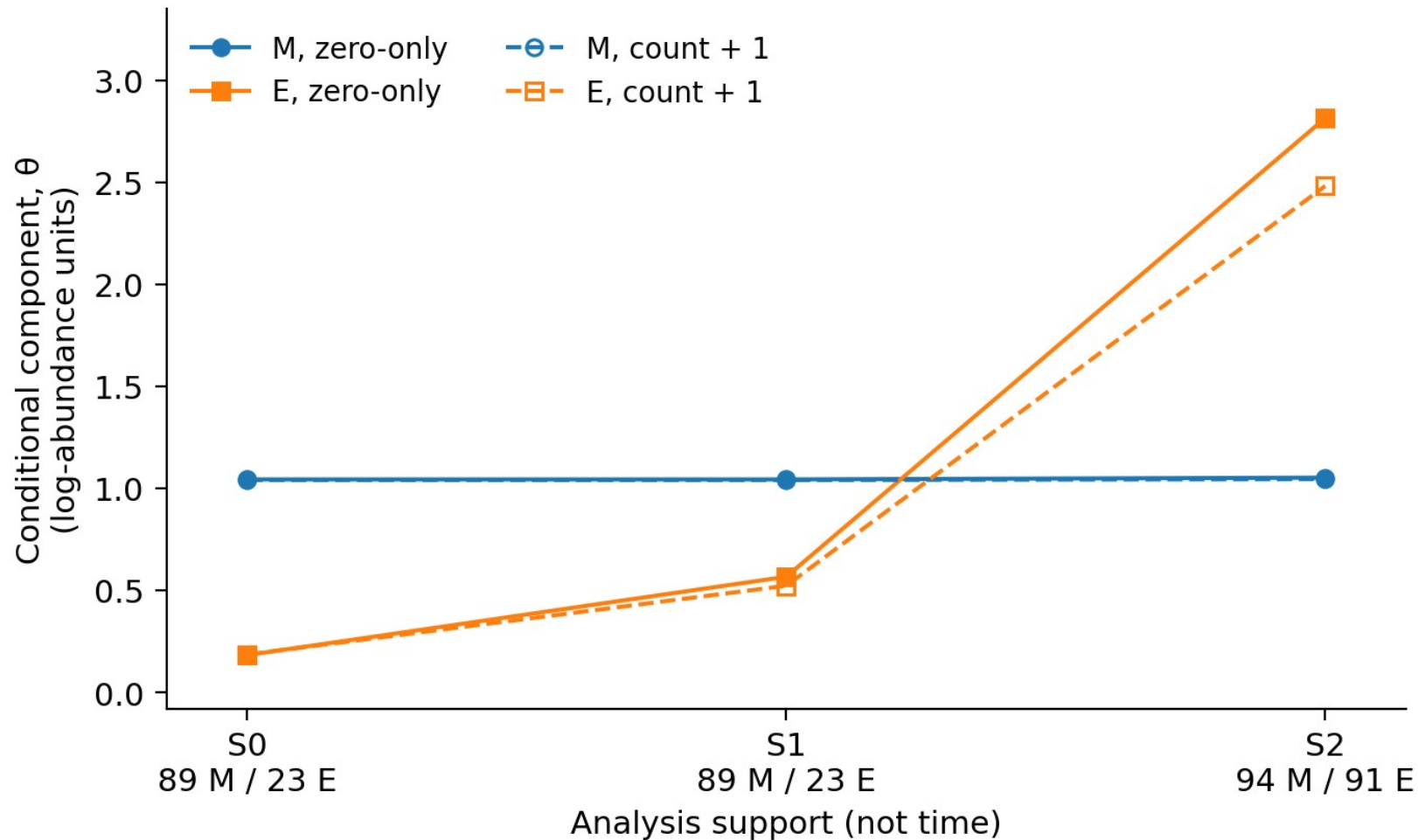
One experiment; three analysis populations

16 sample records: four endpoint groups × 3, plus two T0 records per background.

S0	Original genes and targeting units	M 89 / E 23
S1	Same genes; expanded targeting units	M 89 / E 23
S2	All T0-qualified support	M 94 / E 91

M: Mitochondrial translation. E: Eukaryotic Translation Elongation.

The comparator component changes

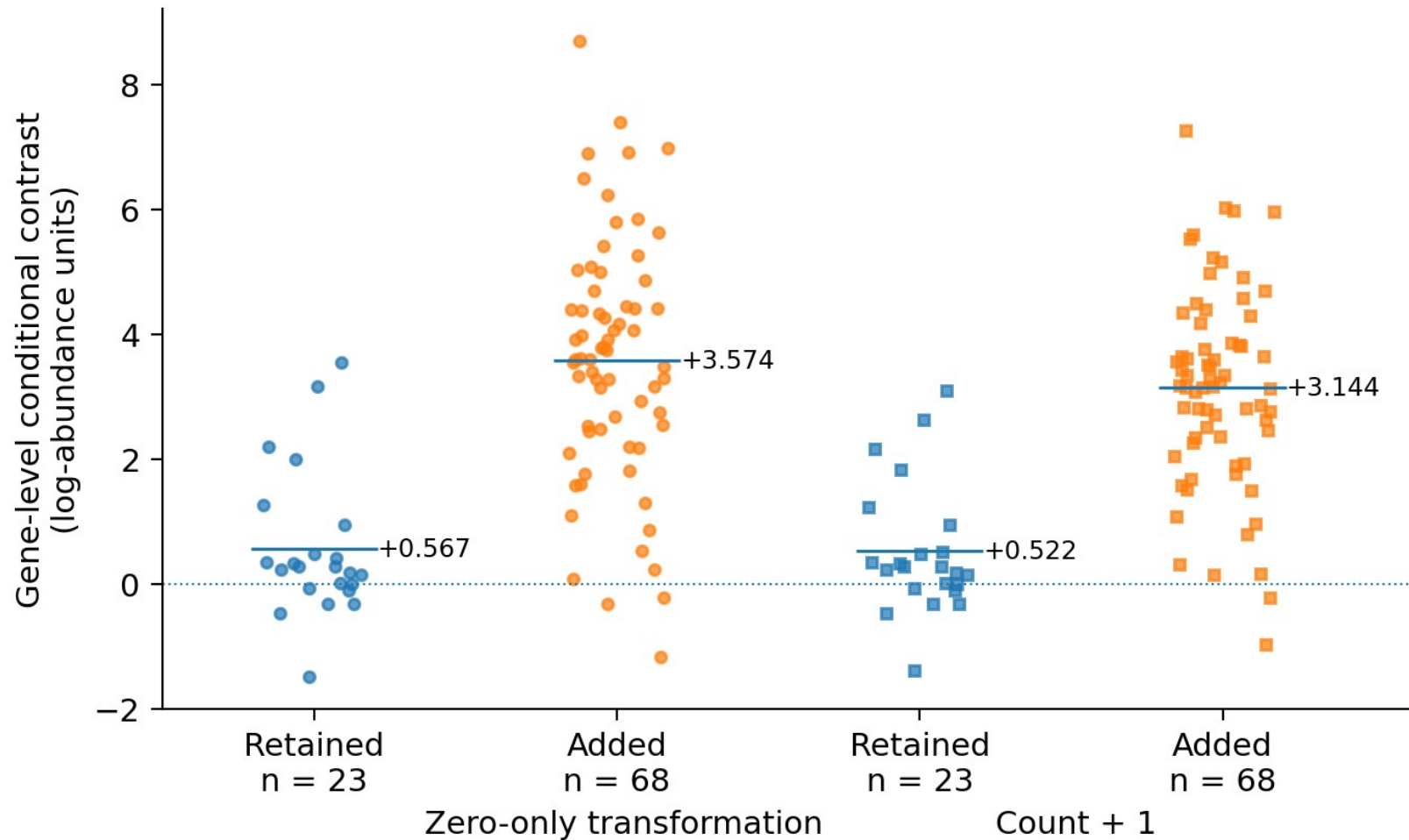


M is nearly stable.

E increases with restored support.

Both transforms show this pattern.

Added comparator genes shift the mixture



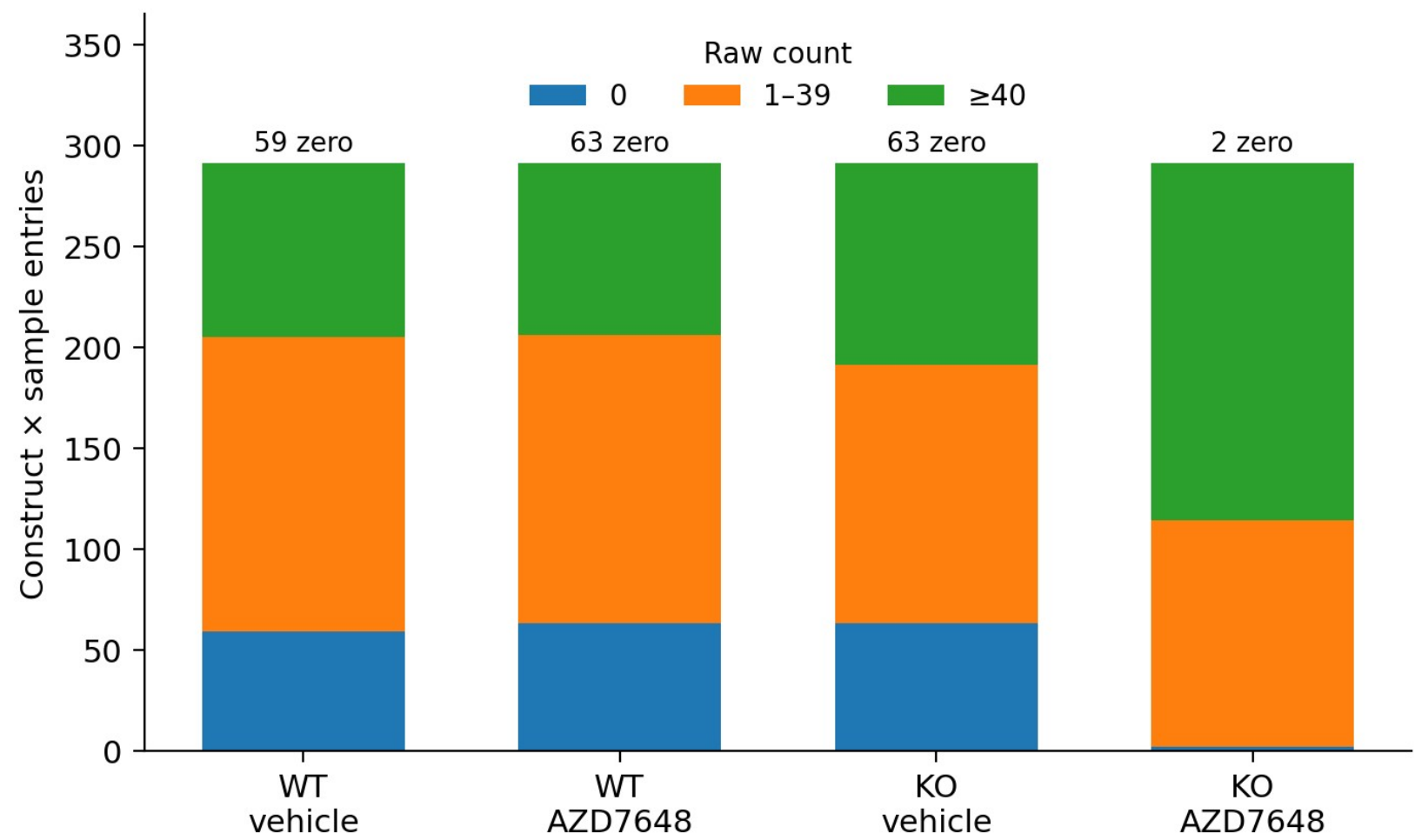
Zero-only means

23 retained:
+0.567

68 added: +3.574

All members
retained in the
display.

More coverage is not automatically better evidence



Qualification uses vehicle.

The response subtracts vehicle.

Coverage, growth and count floors remain entangled.

A finite conclusion, not a target mechanism

The original positive M – E contrast does not provide a membership-insensitive basis for a mitochondrial-specific priority.

The expanded negative contrast is not biological truth.

Clone, growth and perturbation-specific effects are not isolated.

Earlier unsuccessful comparisons remain separate case histories.

Contribution and reproducible scope

Project: fixed comparisons, component decomposition, source-linked code.

Original investigators: laboratory experiments and published mechanisms.

AI assistance: discussion, implementation, testing, figures and writing.

Public core: transformed gene/sample records → pathway summaries → six comparisons.

Historical source tables require separate retrieval.

Backup: the quantity being compared

$$\beta(P, b) = \text{mean drug}(P, b) - \text{mean vehicle}(P, b)$$

$$\theta(P) = \beta(P, KO) - \beta(P, WT)$$

$$\text{Pathway contrast} = \theta(M) - \theta(E)$$

A common additive center cancels.

Membership, count floors and clone effects do not.

Backup: inspect, reproduce, then interpret

Report: reports/DDR_TargetBridge_Case_Study.pdf

Claims: docs/SCIENTIFIC_CLAIM_LEDGER.md

Verify: python tools/run.py verify

Public core: python tools/run.py core --output .build/demo

Historical tables: python tools/run.py historical --output .build/history

github.com/jiangboyan0129/DDR-TargetBridge